MOCK - Tracing a Small Program
SYNTHETIC LOCAL DEMO - NOT INSTITUTIONAL CONTENT OR APPROVAL

Objective: Trace an input, two sequential assignments and an output, preserving the order of updates.

MOCK ORIGINAL LESSON. A variable is a named place for a current value. Input supplies a value, processing changes or combines values, and output reports a result. Read instructions from top to bottom. An assignment replaces the old value; it does not keep every earlier value.

WORKED EXAMPLE
Start with count = 2. First run count = count + 3, so count becomes 5. Then run count = count * 2, so count becomes 10. Displaying count outputs 10. Reversing the two assignments would produce 7, so their order matters.

CHECK AND CORRECTION
Which instruction is processing rather than input or output?
Multiplication transforms the supplied values. Reading and displaying belong to the input and output stages.

Before continuing, separate the current value from the instruction. If count is 4, the instruction count = count + 2 replaces it with 6. Only the next instruction sees that updated 6. Writing the value after every line prevents using the original input twice.

INDEPENDENT APPLICATION
Start count = 4. Run count = count + 2, then count = count * 3. What value is displayed?
The addition gives 6; the next line multiplies that 6 by 3 to give 18.

GLOSSARY
Input: A value supplied to a program before a calculation uses it.
Variable: A name associated with a current value that assignments may replace.
Assignment: An instruction that stores a calculated or supplied value in a variable.
Output: A value or message that the program reports.

To trace a sequence independently, record each new value in order, then report the last value at the output instruction. Practice results here are demonstration study data, not a course grade.

Original mock fixture. No real learner data, school policy, signatures or redistribution review.
Version 1. Accessible lesson text and authored checks are included separately in this package.